

Personnel Qualifications Guide

The purpose of this document is to help Government stakeholders from requirement and contracting offices develop their personnel qualification requirements in support of SeaPort-NxG solicitations issued by NAVSEA affiliated contracting activities. This is just a guide to help create a tailorable attachment to include on a solicitation. Government stakeholders shall tailor their personnel qualification to meet requirement owner's needs in a separate document/spreadsheet and include it as an attachment to the solicitation.

When the government solicits contractor support services, they typically have wide latitude in expressing how their personnel qualification requirements are conveyed in a way that industry must comply with in their proposal; however, they are typically described in either general or specific terms, or somewhere in the middle of the spectrum.

Characteristics of a general approach:

- Government allows wide latitude for the Offeror to propose a team to complete the requirements. For example, "The contractor shall propose a team with the necessary skills, qualifications, and experience to perform the required tasks.
- Government describes level of effort required across a labor mix that is defined in broad categories (e.g. junior, mid-level, senior).

Characteristics of a specific approach:

- Government is highly specific about the team composition. For example, "The contractor shall provide a team of five individuals consisting of one project manager, one technical lead, two subject matter experts, and one administrative support staff member."
- Government describes level of effort required across a labor mix that is defined in highly exact terms (e.g. Government defined labor categories).
- Government sometimes goes into even more specifics about the labor category qualifications, education levels, years of relevant experience, performance location, and so on.

Below considerations should be taken into account when deciding between the general and specific approaches:

- Greater control over the project: Specifying the labor mix can provide greater control over the project and ensure that the required skill sets are present. It can also help to ensure that the team has the right balance of experience, knowledge, and skills to meet the agency's needs.
- Improved accountability: When the government specifies the labor mix, the industry must comply with it, which can improve accountability. This can also help to reduce the risk of the contractor using less experienced or less qualified personnel to reduce costs.
- Reduced risk of disputes: Specifying the labor mix upfront can help to reduce the risk of disputes later in the project. By establishing clear expectations and requirements, the government can ensure that the contractor meets the necessary qualifications, which can reduce the likelihood of problems arising during the project.
- Reduced flexibility: Specifying the labor mix can reduce the flexibility of the contractor to adjust the staffing as the project progresses. This can lead to problems if the needs of the project change or if unexpected issues arise.
- Higher costs: If the government specifies the exact labor mix, it may limit competition, which can lead to higher costs. This is because contractors who cannot meet the required labor mix may not bid on the project, reducing the number of available options.
- Potential for noncompliance: If the government specifies the labor mix, it is essential to ensure that the contractor complies with the requirements. If the contractor does not meet the specified labor mix, it can lead to issues with the project and potentially delay the work.

In summary, while specifying the labor mix can provide greater control and accountability, it can also limit flexibility and potentially lead to higher costs. It is essential to carefully consider the pros and cons before deciding whether to use this approach in government solicitations for professional support services. Ultimately, the decision to use specific or general labor mix language will depend on the needs of the requiring office and the nature of the project.

Specific Approach:

Offerors' proposals may not deviate from the total number of hours included in Section B.

Offerors have the latitude to propose the labor mix they believe is best to accomplish the requirements identified within the SOW in a high quality, efficient, and cost effective manner. Nevertheless, the Government has provided the below labor mix estimate to serve as a reference for prospective Offerors to better understand the overall distribution of skills throughout this effort. This estimate is based on historical requirements; as such, this estimate will be considered in evaluating proposals. The Offeror may deviate but such deviation shall be explained in the Offeror's Staffing Plan Narrative. If the proposed labor mix differs from the estimate provided above, the Offeror shall describe how/why the proposed labor mix will result in a high quality, efficient, and cost effective solution. If the Offeror does not explain deviations from the above labor mix it may result in upward cost adjustments or one or more weaknesses/deficiencies during evaluation.

For each of the Non-Surge Labor CLINs, the Government strongly recommends offerors provide the following labor mix, which includes all Key Personnel:

Labor Category	CLIN 6000, 6100, 6200, 6300 Annual Labor Hours	6400 Annual Labor Hours
Key Personnel (All FTEs)	30,912	30,912
Senior Computer Systems/Software Engineer	0	1,920
Systems Engineer	5,760	5,760
Principal Systems Engineer	3,840	3,840
Electrical Engineer	0	2,112
Mechanical Engineer	5,760	5,760
Computer System/Software Engineer	3,840	3,840
Software Developer	7,680	7,680
Software Developer (Threat Engineering)	7,680	7,680
Junior Engineer	13,824	13,824
Configuration Management Analyst	3,840	3,840
Program Analyst	24,000	24,000
Senior Writer/Editor	1,920	1,920
Engineering Technician III – 30083	0	1,056
Engineering Technician IV – 30084	0	1,056
Engineering Technician V -- 30085	7,872	7,872
Engineering Technician VI – 30086	0	1,056
Windows System Admin	7,680	7,680
Linux System Admin	7,680	7,680
IA Specialist (Cyber Lethality Support)	5,760	5,760
Technical Specialist	1,920	1,920
Management & Support	2,400	2,400
Total	138,720	149,088

A. Government Defined Labor Categories:**SENIOR COMPUTER SYSTEMS/SOFTWARE ENGINEER**

A Bachelor's degree from an accredited college or university in an engineering, scientific, or technical discipline. Ten (10) years professional technical experience which includes at least five (5) years specialized experience performing computer hardware engineering, software design, coding and compilation of code; computer or software

requirements management, computer systems or software integration; and utilizing systems engineering software tools. Specialized experience also includes demonstrated ability to engineer complex systems with integrated software and hardware solutions. Experience related to weapons systems or Navy programs and Hypertext Markup Language (HTML), C/C++, Unix, and Linux. Specialized experience with development of Real-Time VxWorks Symmetric Multi-Processing (SMP) software. Expertise related to hardware and software development for Standard Missile (SM)-2, SM-3, SM-6, and Evolved Sea Sparrow Missile (ESSM) missile internal interfaces, such as experience with SM-3 Kill Warhead Data Link (KWDL), SM-3 Missile Test Data (MTD), SM-2/Sm-6 initialization message digital interface, and vertical launch system.

SYSTEMS ENGINEER

A Bachelor's degree from an accredited college or university in an engineering, scientific, or technical discipline. Five (5) years professional technical experience which includes at least three (3) years specialized experience performing systems requirements management, systems engineering design, utilizing systems engineering software tools and performing systems engineering design, integration engineering, test and evaluation engineering activities.

PRINCIPAL SYSTEMS ENGINEER

A Bachelor's degree from an accredited college or university in an engineering, scientific, or technical discipline. Eight (8) years professional technical experience including at least five (5) years specialized experience in the areas of maritime, launcher, missile, weapon, or weapon control systems concept definition, design, integration, and engineering. Specialized experience also includes systems requirements management, systems engineering design, utilization of systems engineering software tools, and systems engineering processes. Engineering experience in developing requirements, risk management, and requirements traceability and verification methods associated with weapon systems, specifically missiles, launchers, or weapons control systems. Engineering experience in the design, development, and testing of surface launching system components for reliability. Engineering experience in generating system architecture, all levels of system requirements, interface diagrams, test matrix, and CONOPS. Experience in providing information for system engineering events.

ELECTRICAL ENGINEER

A Bachelor's degree from an accredited college or university in an engineering, scientific, or technical discipline. Five (5) years professional technical experience which includes three (3) years specialized experience performing design and integration engineering, test and evaluation activities, electrical systems requirements management, electrical systems engineering design, utilization of electrical systems engineering design software tools, and electrical systems engineering processes.

MECHANICAL ENGINEER

A Bachelor's degree from an accredited college or university in an engineering, scientific, or technical discipline. Five (5) years professional technical experience which includes three (3) years specialized experience that includes performing mechanical systems requirements management, mechanical systems engineering design and integration, utilizing mechanical systems engineering design software tools (such as SolidWorks or ProEngineer), and mechanical systems engineering processes.

COMPUTER SYSTEMS/SOFTWARE ENGINEER

A Bachelor's degree from an accredited college or university in an engineering, scientific, or technical discipline. Five (5) years professional technical experience which includes at least three (3) years specialized experience performing computer hardware engineering, software design, coding and compilation of code; computer or software requirements management, computer systems or software integration; and utilizing systems engineering software tools. Specialized experience also includes demonstrated ability to engineer complex systems with integrated software and hardware solutions. Experience related to weapons systems or Navy programs and HTML, C/C++, Unix, and Linux is desired.

SOFTWARE DEVELOPER

A Bachelor's degree from an accredited college or university in an engineering, scientific, or technical discipline, or four (4) years of a demonstrated and established knowledge base and subject matter expertise relevant to specific technical disciplines of the SOW. In addition, ten (10) years of experience as a software developer creating or maintaining operating systems, communications software, database packages, compilers, assemblers or utility programs for software applications. Seven (7) years of experience with multiple programming languages such as C, C++, C#, Java, DotNET, Ruby, Erlang, HTML5, JavaScript, JQuery, Ajax and Rails web development framework, Voice over IP (VoIP) and Session Initiated Protocol (SIP), and with software development lifecycle, architecture, and design experience. Experience with Unified Modeling Language (UML), Microsoft (MS) Project, and MS Office tools. Experience in the analysis and interpretation of user requirements and in the development of design documentation. Experience in designing, coding, benchmark testing, debugging, and documentation of software programs. Demonstrated ability to develop and support technical positions, to justify technical recommendations, and to write technical publications and reports.

SOFTWARE DEVELOPER (THREAT ENGINEERING)

A Bachelor's degree from an accredited college or university in an engineering, scientific, or technical discipline, or four (4) years of a demonstrated and established knowledge base and subject matter expertise relevant to specific technical disciplines of the SOW. Specialized experience with Next Generation Threat System (NGTS) and Advanced Framework for Simulation Integration and Modeling (AFSIM) is desired to support the assessment of Blue Force weapons and tactics in simulated engagements and to augment weapons testing. Experience with combat management systems and weapons systems, particularly missiles, is highly desired. Technical skills and abilities should include experience with modern C++ programming (e.g., C++11 or later), experience with multiple simulation environments, particularly NGTS, AFSIM, or the Integrated Threat Analysis and Simulation Environment (ITASE), familiarity with Agile methodologies (e.g., Scrum, Kanban), and proficiency in using debugging tools and code profiling.

JUNIOR ENGINEER

A Bachelor's degree from an accredited college or university in an engineering, scientific, or technical discipline. Requires knowledge of principles and practices of mechanical engineering, electrical engineering, or systems engineering.

CONFIGURATION MANAGEMENT ANALYST

Four (4) years general experience which includes three (3) years specialized experience in configuration management demonstrating a thorough and comprehensive knowledge of configuration management principles, methods, and techniques. Experience evaluating change documentation and developing impact assessments for planned configuration changes; maintaining integrity and traceability of configurations throughout its life cycles; experience with formal Change Control Boards related to Engineering Change Proposals (ECPs), Specification Change Notices (SCNs), Notice of Revisions (NORs), Deviations, or Waivers.

PROGRAM ANALYST

Three to Five (3-5) years general experience which includes four (4) years specialized experience in analyzing and evaluating the effectiveness of program operations in meeting established goals, requirements, and objectives; assisting in developing Life-Cycle cost analyses of projects or performing cost benefit or economic evaluations of current or projected programs; analyzing information to develop program reporting systems including the systems specifications, data gathering and analytical techniques, and systems evaluation methodology; analyzing new or proposed legislation, regulations, or requirements to determine impact on program operations, engineering design, or security; assisting in developing management or program evaluation plans, procedures, and methodology; analyzing and evaluating proposed changes in mission or operating procedures. Experience related to weapons systems or Navy programs.

SENIOR WRITER/EDITOR

Seven (7) years of technical editing experience. Experience researching events and data and conducting interviews of individuals in attendance at an event. Experience writing, editing, and obtaining release approvals for articles, feature stories, news releases, speeches, briefings, scripts, and statements, including electronic, web-based, and hard-copy material. Experience collecting and reviewing presentation materials, including electronic and web-based material. Experience reviewing and formatting technical documentation such as test plans, reports, specifications, white papers, power point presentations, and system engineering program reviews.

ENGINEERING TECHNICIAN III - 30083

Five (5) years of experience in applying engineering services to maritime, launcher, missile, weapon, or weapon control system components and assemblies. This experience includes developing and testing solutions to solve technical problems in research and development, manufacturing, inspection, and maintenance, logistics support, and assisting engineers in inspecting products, conducting tests, and collecting data.

ENGINEERING TECHNICIAN IV – 30084

Seven (7) years of experience in applying engineering services to maritime, launcher, missile, weapon, or weapon control system components and assemblies. This experience includes developing and testing solutions to solve technical problems in research and development, manufacturing, inspection, and maintenance, logistics support, and assisting engineers in inspecting products, conducting tests, and collecting data.

ENGINEERING TECHNICIAN V – 30085

Ten (10) years of experience in applying engineering services to maritime, launcher, missile, weapon, or weapon control system components and assemblies. This experience includes developing and testing solutions to solve technical problems in research and development, manufacturing, inspection, and maintenance, and assisting engineers in inspecting products, conducting tests, and collecting data.

ENGINEERING TECHNICIAN VI – 30086

Fifteen (15) years of experience in applying engineering services to maritime, launcher, missile, weapon, or weapon control system components and assemblies. This experience includes developing and testing solutions to solve technical problems in research and development, manufacturing, inspection, and maintenance, and assisting engineers in inspecting products, conducting tests, and collecting data.

SYSTEMS ADMINISTRATOR (UNIX/LINUX)

Three (3) years of professional experience in Unix and/or Linux system administration of networks and software systems. Experience in system backup and recovery; basic software management; security management and library management; installation and configuration of hardware; operational systems and Commercial Off the Shelf (COTS) software; operating computer systems in different operating environments; and sizing, performance, tuning, hardware/software/network trouble shooting and resource allocation. Qualified experience in different network protocols including Transmission Control Protocol/Internet Protocol (TCP/IP) and major network security tools such as: firewalls, knowledge of computing and networking vulnerabilities, attack methods and latest detection methods. Per DoD Directive 8570.01-M or successor, this position requires Information Assurance Technical (IAT) Level 2 or higher and requires this position to be designated as an Information Technology (IT) Level 1.

SYSTEMS ADMINISTRATOR (WINDOWS)

Three (3) years of professional experience in relation to Windows system administration of networks and software systems. Experience in system backup/recovery, basic software management, security and library management, installation and configuration of hardware, operating systems and COTS software, operating computer systems in different operating environments, sizing, performance, tuning, hardware/software /network trouble shooting and resource allocation. Experience in maintaining proficiency on major network security tools such as, firewalls. Experience developing system specifications and performing market research procurement tasks. Qualified experience with different network protocols including Transmission Control Protocol/Internet Protocol (TCP/IP). Qualified experience of computing and networking vulnerabilities, and attack and latest detect methods. Per DoD Directive 8570.01-M or successor, this position requires IAT Level 2 or higher and requires this position to be designated as an IT Level 1.

INFORMATION ASSURANCE SPECIALIST (CYBER LETHALITY SUPPORT)

A Bachelor's degree from an accredited college or university in an engineering, scientific, business, or technical discipline is desired. Five (5) years professional experience in Information Assurance that includes two (2) years performing independent information technology product assurance activities in support of the design, development, and implementation of automated systems, and enhancement of existing systems. Experience in the management and administration of IT systems and dissemination of critical mission information across multiple domains and operational venues. Experience directing the efforts to obtain the certification and accreditation of the military systems via the Risk Management Framework (RMF), Defense Information Assurance Certification and

Accreditation Process (DIACAP), Platform Information Technology (PIT) or predecessor processes. Per DoD Directive DoDI 8570.01-M or successor, this position requires a Information Assurance Manager (IAM) Level II certification(s).

TECHNICAL SPECIALIST

Three (3) years of experience with the design, implementation, manufacturing, configuration, and check out of networks, systems, computer components, peripheral equipment, or networking hardware; with knowledge of computer processors, displays, input/output devices, and hardware interfaces. Experience must include at least one (1) of the below areas:

Experience using engineering analysis and development tools such as Labview and Matlab. Experience with upkeep of an electromechanical laboratory. Experience building and testing cables and electronic sub-assemblies. Experience in the machining, assembly or testing of prototype mechanical devices from provided designs. Experience with CAD software such as SolidWorks/Pro Engineer. Experience with systems analysis, technical documentation, maintenance, repair, operation, and support of system or system components. Desired qualification: Operator Proficiency Certified for J-STD-001 (J-STD-001 is a soldering certification for electrical and electronic assembly manufacturing and is considered an industry standard). IPC-A-610 certification (Acceptability of Electronic Assemblies).

B. Geographic Distribution

For proposal purposes, Offerors should assume that all of the support will be provided in accordance with the requirements as outlined in section C of the solicitation.

C. Key Personnel Qualifications

The solicitation requires eight (8) Key Personnel positions, with Program Manager and Senior Software Engineer each requiring 1 FTE to fill; Senior Mechanical Engineer, Principal Engineer, Electrical Engineer – Junior, and Electrical Engineer Mid/Journeyman each will require 2 FTEs to fill; Senior Systems Engineer will require 2.5 FTE to fill; Senior Electrical Engineer will require 3 FTEs to fill. All Key Personnel positions shall be considered senior. The following table lists these Key Personnel positions, as well as the Government's desired attributes for each:

Key Personnel Position	Desired Attributes
Program Manager	• Bachelor's degree from an accredited college or university in an engineering, scientific, business, or technical discipline • Fifteen (15) years professional technical experience including at least five (5) years specialized experience in the management of the development or evaluation of Department of Defense (DoD) Weapons Systems • Ten (10) years program management experience, which includes both technical and administrative aspects of Government contract performance, providing contract data deliverables, and having oversight of contract schedule and cost • Experience related to maritime, launcher, missile, or DOD weapon control or effectiveness systems.

Senior Systems Engineer

- Bachelor's degree from an accredited college or university in an engineering, scientific, or technical discipline
- Ten (10) years professional technical experience including at least five (5) years specialized experience in the areas of maritime, launcher, missile, weapon, or weapon control systems concept definition, design, integration, and engineering
- Specialized experience also includes systems requirements management, systems engineering design, utilization of systems engineering software tools, and systems engineering processes
- Engineering experience in developing requirements, risk management, and requirements traceability and verification methods associated with weapon systems, specifically missiles, launchers, or weapons control systems
- Engineering experience in the design, development, and testing of surface launching system components for reliability
- Engineering experience in generating system architecture, all levels of system requirements, interface diagrams, test matrix, and Concept of Operations (CONOPS)
- Experience in providing information for system engineering events

Senior Mechanical Engineer

- Bachelor's degree from an accredited college or university in an engineering, scientific, or technical discipline
- Ten (10) years professional technical experience including at least five (5) years specialized experience in the areas of maritime, launcher, missile, weapon, or weapon control systems concept definition, design, integration, and engineering
- Specialized experience also includes experience as a mechanical engineer performing mechanical engineering analysis and design, utilizing mechanical engineering software tools, coordinating mechanical fabrication, and mechanical engineering test techniques
- Engineering experience in the mechanical design, development, and testing of missile launcher and support equipment
- Experience with structural design and analysis, finite element analysis, fluid dynamics, and thermal dynamics
- Experience with SolidWorks, LS-DYNA, or Craft Tech software

Senior Software Engineer	• Bachelor's degree in Computer Science or Mathematics obtained from an accredited college or university is required • Ten (10) years of experience as a skilled developer creating or maintaining operating systems, communications software, database packages, compilers, assemblers, and utility programs for software applications • Experience programming at the field-programmable gate array (FPGA) and microcontroller levels is desired • Experience in biometrics, surveillance, sensors, or cyber-security is desired • Seven (7) years of experience with multiple programming languages including C, C++, C#, Java, DotNET and with software development Life-Cycle, architecture, and design is desired • Experience in both Windows and Unix development is also desired • Experience with Unified Modeling Language (UML), Microsoft (MS) Project and MS Office tools • Experience in designing, coding, benchmark testing, debugging, and documentation of software programs • Demonstrated ability to develop and support technical positions, to justify technical recommendations, and to write technical publications and reports
Senior Electrical Engineer	• Bachelor's degree from an accredited college or university in an engineering, scientific, or technical discipline • Ten (10) years professional technical experience including at least five (5) years specialized experience in the areas of maritime, launcher, missile, weapon, or weapon control systems concept definition, design, integration, and engineering • Specialized experience also includes experience as an electrical engineer performing electrical engineering analysis and design, in electrical engineering software tools, coordinating electrical fabrication, and electrical engineering test techniques • Engineering experience in the design and integration of sensors (tracking and guidance) into projectiles and weapon systems • Engineering experience in the design, development, and testing of missile launcher electrical equipment • Experience with power systems, embedded controllers, and linear drive systems

Principal Engineer	• Twenty (20) years of missile integration and test experience with at least seven (7) years of specialized experience ◦ Specialized experience is defined as significant systems engineering responsibilities in a minimum of two of the following areas: a. Missile Systems; b. Weapon/Combat Systems; c. Aegis Ballistic Missile Defense; d. Experience with Interface and interactions between Aegis Weapon System (AWS) computer program, and the Vertical Launching System (VLS) during Standard Missile prelaunch sequencing, beacon capture and through midcourse guidance phase of flight and transition to terminal homing; e. Radar and Radio Frequency (RF) Communications Experience; f. Weapon/Combat Systems/Missile Systems Electrical and Software Interfaces; g. Design and Development of High-Performance Embedded Systems • This position requires extensive technical leadership experience, and is responsible for guiding the technical vision, strategy, and execution of major engineering projects and initiatives within an organization • They serve as a subject matter expert, provide technical leadership and mentorship to engineering teams, oversee high-level design and architecture, and collaborate with both executive management and technical staff to ensure projects align with strategic goals and are delivered efficiently and to high standards
Electrical Engineer - Junior	• Bachelor's Degree in Electrical Engineering plus three (3) years of experience in electrical design and engineering techniques • Demonstrated ability to evaluate electrical systems, products, components, and applications by designing and conducting research programs; applying knowledge of electricity and materials for electrical systems • Demonstrated ability to devise new approaches to problems, prepare or modify drawings and specifications, and provide quality control of projects built by others. • Experience with RF/microwave test equipment • Demonstrated ability to perform analog circuit/network analysis, and an understanding of power semiconductor physics

Electrical Engineer-
Mid/Journeyman

- Bachelor's Degree in Electrical Engineering plus three (3) years of experience in electrical design and using engineering techniques
- Demonstrated ability to evaluate electrical systems, products, components, and applications by designing and conducting research programs; applying knowledge of electricity and materials for electrical systems
- Demonstrated ability to devise new approaches to problems, and prepare or modify drawings, specifications, and provide quality control of projects built by others
- Experience with RF/microwave test equipment
- Demonstrated ability to perform analog circuit/network analysis, and an understanding of power semiconductor physics

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